Chapter 11. The Analysis of Covariance (ANACOVA)

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1 Introduction

- ANACOVA combines the features of ANOVA and regression.
- It reduces a large error variance in the ANOVA model by adding one or more quantitative variables (covariates) linearly related to the response.
- Setting: besides the response y there is a covariate x, linearly related to y, that *cannot be controlled* by the experimenter but can be *observed* along with y (recall Ch. 4: known but uncontrollable nuisance factor).

Example 11.1: Cracker promotion

A company studies three promotion types on cracker sales. 15 stores; each store is randomly assigned to one promotion (5 stores per type). y = number of cases sold during the promotional period; x = sales in the preceding period (covariate).

	Store 1		2		3		4		5	
Trt	y	x	y	x	y	x	y	x	y	x
1	38	21	39	26	36	22	45	28	33	19
2	43	34	38	26	38	29	27	18	34	25
3	24	23	32	29	31	30	21	16	28	29

The scatter of y around the treatment averages is large, but the scatter around the *treatment regression lines* (on x) is much smaller: part of the "error" is explained by x.

2 Model

Definition 11.2

$$y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}, \qquad i = 1, \ldots, a;\ j = 1, \ldots, n,$$

x_{ij} given (fixed), $\sum_i \tau_i = 0$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, β = common regression coefficient (slope). Hence

$$\mathrm{E}(y_{ij}) = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}), \qquad \mathrm{Var}(y_{ij}) = \sigma^2.$$

Each treatment has its own regression line, all **parallel** with slope β; τ_i is the vertical shift.

2.1 Notation: corrected sums of squares and cross products

$$S_{xx} = \sum_i \sum_j (x_{ij} - \bar{x}_{\cdot\cdot})^2, \quad S_{xy} = \sum_i \sum_j (x_{ij} - \bar{x}_{\cdot\cdot})(y_{ij} - \bar{y}_{\cdot\cdot}), \quad S_{yy} = \sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2,$$
$$T_{xx} = n \sum_i (\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2, \quad T_{xy} = n \sum_i (\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}), \quad T_{yy} = n \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2,$$
$$E_{xx} = \sum_i \sum_j (x_{ij} - \bar{x}_{i\cdot})^2, \quad E_{xy} = \sum_i \sum_j (x_{ij} - \bar{x}_{i\cdot})(y_{ij} - \bar{y}_{i\cdot}), \quad E_{yy} = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot})^2,$$

with $S = T + E$ in each case (same proof as the one-way ANOVA identity, applied to products).

3 Least Squares Estimation

Motivation. With a covariate, treatment means must be compared at a common covariate value. Analysis of covariance fits parallel regression lines and compares their intercepts, which removes covariate variation from the error.

Derivation

$S = \sum_i \sum_j \left(y_{ij} - \mu - \tau_i - \beta(x_{ij} - \bar{x}_{\cdot\cdot})\right)^2$.

$$\frac{\partial S}{\partial \mu} = 0 \Rightarrow \sum_i \sum_j \left(y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}(x_{ij} - \bar{x}_{\cdot\cdot})\right) = 0 \Rightarrow \hat{\mu} = \bar{y}_{\cdot\cdot},$$
$$\frac{\partial S}{\partial \tau_i} = 0 \Rightarrow n\hat{\mu} + n\hat{\tau}_i + \hat{\beta} \sum_{j=1}^{n} (x_{ij} - \bar{x}_{\cdot\cdot}) = y_{i\cdot} \Rightarrow \hat{\tau}_i = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) - \hat{\beta}(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}),$$
$$\frac{\partial S}{\partial \beta} = 0 \Rightarrow \sum_i \sum_j \left(y_{ij} - \hat{\mu} - \hat{\tau}_i - \hat{\beta}(x_{ij} - \bar{x}_{\cdot\cdot})\right)(x_{ij} - \bar{x}_{\cdot\cdot}) = 0.$$

Theorem 11.3

$$\hat{\beta} = \frac{E_{xy}}{E_{xx}}, \qquad \hat{\mu} = \bar{y}_{\cdot\cdot}, \qquad \hat{\tau}_i = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) - \hat{\beta}(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot}).$$

Proof [Supplement]. (In $\partial S/\partial\mu$, $\sum \hat{\tau}_i = 0$ and $\sum\sum(x_{ij} - \bar{x}_{\cdot\cdot}) = 0$ give $N\hat{\mu} = y_{\cdot\cdot}$...) Expand the β-equation:

$$\sum\sum y_{ij}(x_{ij} - \bar{x}_{\cdot\cdot}) - \hat{\mu} \underbrace{\sum\sum (x_{ij} - \bar{x}_{\cdot\cdot})}_{=0} - \sum_i \hat{\tau}_i \sum_j (x_{ij} - \bar{x}_{\cdot\cdot}) - \hat{\beta} \sum\sum (x_{ij} - \bar{x}_{\cdot\cdot})^2 = 0.$$

Since $\sum\sum y_{ij}(x_{ij}-\bar{x}_{\cdot\cdot}) = S_{xy}$ and $\sum_j (x_{ij}-\bar{x}_{\cdot\cdot}) = n(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot})$,

$$\sum_i \hat{\tau}_i\, n(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot}) + \hat{\beta} S_{xx} = S_{xy}.$$

Substitute $\hat{\tau}_i$:

$$\underbrace{n\sum_i (\bar{y}_{i\cdot}-\bar{y}_{\cdot\cdot})(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot})}_{=T_{xy}} - \hat{\beta}\underbrace{n\sum_i (\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot})^2}_{=T_{xx}} + \hat{\beta} S_{xx} = S_{xy}$$

$$\Rightarrow \hat{\beta}(S_{xx}-T_{xx}) = S_{xy}-T_{xy} \;\Rightarrow\; \hat{\beta} = \frac{E_{xy}}{E_{xx}}.$$

So the slope is estimated *within* treatments (pooled within-group regression). ■

Theorem 11.4: Error sum of squares

$$SS_E = E_{yy} - \frac{E_{xy}^2}{E_{xx}}, \qquad \text{df} = a(n-1)-1, \qquad \text{Var}(\hat{\beta}) = \frac{\sigma^2}{E_{xx}}.$$

Proof [Supplement]. Fitted value $\hat{y}_{ij} = \hat{\mu} + \hat{\tau}_i + \hat{\beta}(x_{ij}-\bar{x}_{\cdot\cdot}) = \bar{y}_{i\cdot} + \hat{\beta}(x_{ij}-\bar{x}_{i\cdot})$, so $e_{ij} = (y_{ij}-\bar{y}_{i\cdot}) - \hat{\beta}(x_{ij}-\bar{x}_{i\cdot})$ and

$$SS_E = \sum\sum e_{ij}^2 = E_{yy} - 2\hat{\beta}E_{xy} + \hat{\beta}^2 E_{xx} = E_{yy} - 2\frac{E_{xy}^2}{E_{xx}} + \frac{E_{xy}^2}{E_{xx}} = E_{yy} - \frac{E_{xy}^2}{E_{xx}}.$$

Parameters: μ, $a-1$ free τ's, β $\Rightarrow$ $a+1$; df $= an-a-1 = a(n-1)-1$. Variance: $\hat{\beta} = \sum\sum(x_{ij}-\bar{x}_{i\cdot})y_{ij}/E_{xx}$ is linear in independent y's, so $\text{Var}(\hat{\beta}) = \sigma^2\sum\sum(x_{ij}-\bar{x}_{i\cdot})^2/E_{xx}^2 = \sigma^2/E_{xx}$. ■

Adjusted treatment means: $\hat{\mu}+\hat{\tau}_i = \bar{y}_{i\cdot} - \hat{\beta}(\bar{x}_{i\cdot}-\bar{x}_{\cdot\cdot})$ — the treatment mean adjusted to the common covariate value $\bar{x}_{\cdot\cdot}$.

4 Regression (Matrix) Formulation

Derivation

With $a=3$, $n=5$ and $\tau_3 = -\tau_1-\tau_2$, index the 15 observations by ℓ:

$$y_\ell = \mu + \tau_1 z_{1\ell} + \tau_2 z_{2\ell} + \beta(x_\ell - \bar{x}) + \varepsilon_\ell, \qquad z_1 = \begin{cases} 1 & \text{trt 1} \\ -1 & \text{trt 3} \\ 0 & \text{otherwise} \end{cases}, \; z_2 = \begin{cases} 1 & \text{trt 2} \\ -1 & \text{trt 3} \\ 0 & \text{otherwise} \end{cases},$$

$$\mathbf{X} = \left[\begin{array}{ccc|c} 1 & 1 & 0 & x_{11}-\bar{x}_{\cdot\cdot} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 0 & x_{15}-\bar{x}_{\cdot\cdot} \\ \hline 1 & 0 & 1 & x_{21}-\bar{x}_{\cdot\cdot} \\ \vdots & & & \vdots \\ 1 & 0 & 1 & x_{25}-\bar{x}_{\cdot\cdot} \\ \hline 1 & -1 & -1 & x_{31}-\bar{x}_{\cdot\cdot} \\ \vdots & & & \vdots \\ 1 & -1 & -1 & x_{35}-\bar{x}_{\cdot\cdot} \end{array}\right], \qquad \hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\mu} \\ \hat{\tau}_1 \\ \hat{\tau}_2 \\ \hat{\beta} \end{pmatrix} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y},$$

$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y}$, $\mathbf{e} = (\mathbf{I}-\mathbf{H})\mathbf{y}$, $SS_E = \mathbf{e}^T\mathbf{e} = \mathbf{y}^T(\mathbf{I}-\mathbf{H})\mathbf{y}$.

Source	DF	
Model $(\tau_1,\dots,\tau_{a-1},\beta)$	a	$F_0 = MS_{\text{Model}}/MS_E$
Error	$an-a-1$	
Total	$an-1$	

Cracker example (`lm`): $\hat{\mu} = 33.800$, $\hat{\tau}_1 = 6.017$, $\hat{\tau}_2 = 0.942$, $\hat{\tau}_3 = -6.959$, $\hat{\beta} = 0.899$.

5 Hypothesis Testing via Extra Sums of Squares

Testing of interest:

(i) $H_0 : \tau_1 = \cdots = \tau_a = 0$;

(ii) $H_0 : \beta = 0$.

Derivation

Full: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$; Reduced: $\mathbf{y} = \mathbf{X}_2\boldsymbol{\beta}_2 + \boldsymbol{\varepsilon}$; $H_0 : \boldsymbol{\beta}_1 = \mathbf{0}$:

$$F_0 = \frac{[SS_E(\mathrm{R}) - SS_E(\mathrm{F})]/[df_R - df_F]}{SS_E(\mathrm{F})/df_F} \overset{H_0}{\sim} F_{df_R-df_F,\, df_F}.$$

(i) Treatments. Full $y_{ij} = \mu + \tau_i + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$ ($df_F = an-a-1$); reduced $y_{ij} = \mu + \beta(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$ (simple regression, $df_R = an-2$), whose error SS is $S_{yy} - S_{xy}^2/S_{xx}$:

$$F_0 = \frac{\left[(S_{yy} - \frac{S_{xy}^2}{S_{xx}}) - (E_{yy} - \frac{E_{xy}^2}{E_{xx}})\right]/(a-1)}{SS_E(\mathrm{F})/[a(n-1)-1]} \overset{H_0}{\sim} F_{a-1,\, a(n-1)-1}.$$

(ii) Slope. Reduced $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$ (one-way ANOVA, error SS E_{yy}, df $a(n-1)$):

$$F_0 = \frac{[E_{yy} - SS_E(\mathrm{F})]/1}{SS_E(\mathrm{F})/[a(n-1)-1]} = \frac{E_{xy}^2/E_{xx}}{MS_E} \overset{H_0}{\sim} F_{1,\, a(n-1)-1} \qquad \Big(= t_0^2,\ t_0 = \frac{\hat{\beta}}{\sqrt{MS_E/E_{xx}}}\Big).$$

Example 11.5: Cracker promotion, line by line

Totals: $x_{i\cdot} = 116, 132, 127$ ($x_{\cdot\cdot} = 375$, $\bar{x}_{\cdot\cdot} = 25$); $y_{i\cdot} = 191, 180, 136$ ($y_{\cdot\cdot} = 507$, $\bar{y}_{\cdot\cdot} = 33.8$); $\sum x^2 = 9735$, $\sum y^2 = 17783$, $\sum xy = 12937$.

$$S_{xx} = 9735 - \frac{375^2}{15} = 9735 - 9375 = 360.0,$$
$$S_{yy} = 17783 - \frac{507^2}{15} = 17783 - 17136.6 = 646.4,$$
$$S_{xy} = 12937 - \frac{375 \cdot 507}{15} = 12937 - 12675 = 262.0,$$
$$T_{xx} = \frac{116^2 + 132^2 + 127^2}{5} - 9375 = 26.8,$$
$$T_{yy} = \frac{191^2 + 180^2 + 136^2}{5} - 17136.6 = 338.8,$$

$$T_{xy} = \frac{116(191) + 132(180) + 127(136)}{5} - 12675 = -37.4,$$
$$E_{xx} = 360.0 - 26.8 = 333.2, \quad E_{yy} = 646.4 - 338.8 = 307.6, \quad E_{xy} = 262.0 + 37.4 = 299.4.$$

$$\hat{\beta} = \frac{E_{xy}}{E_{xx}} = \frac{299.4}{333.2} = 0.8986,$$
$$SS_E(\mathrm{F}) = 307.6 - \frac{299.4^2}{333.2} = 307.6 - 269.03 = 38.57 \quad (11 \text{ df}), \qquad MS_E = 3.506,$$
$$SS_E(\mathrm{R}_\tau) = 646.4 - \frac{262^2}{360} = 646.4 - 190.68 = 455.72 \quad (13 \text{ df}).$$

Treatments: $F_0 = \dfrac{(455.72 - 38.57)/2}{38.57/11} = \dfrac{208.58}{3.506} = 59.5 > F_{0.05,2,11} = 3.98$: reject; promotions differ (R: 59.42).

Slope: reduced model error SS $= E_{yy} = 307.6$ (12 df); $F_0 = \dfrac{(307.6 - 38.57)/1}{3.506} = 76.7 > F_{0.05,1,11} = 4.84$: reject $H_0 : \beta = 0$; the covariate matters. $se(\hat{\beta}) = \sqrt{3.506/333.2} = 0.103$.
Estimates: $\bar{x}_{i\cdot} = 23.2,\ 26.4,\ 25.4$; $\bar{y}_{i\cdot} = 38.2,\ 36.0,\ 27.2$:

$$\hat{\tau}_1 = (38.2 - 33.8) - 0.8986(23.2 - 25) = 4.4 + 1.617 = 6.017,$$
$$\hat{\tau}_2 = (36.0 - 33.8) - 0.8986(26.4 - 25) = 2.2 - 1.258 = 0.942,$$
$$\hat{\tau}_3 = (27.2 - 33.8) - 0.8986(25.4 - 25) = -6.6 - 0.359 = -6.959.$$

Adjusted means: 39.82, 34.74, 26.84. Compare with the one-way ANOVA ignoring x: $MS_E = 307.6/12 = 25.63$ vs 3.51 with the covariate — the error variance is reduced by a factor of 7.

6 Testing for Parallel Slopes

Derivation

Model with treatment-specific slopes:

$$y_{ij} = \mu + \tau_i + \beta_i(x_{ij} - \bar{x}_{\cdot\cdot}) + \varepsilon_{ij}$$

$$\Rightarrow\ y_\ell = \mu + \tau_1 z_{1\ell} + \tau_2 z_{2\ell} + \beta_1(x_{1\ell} - \bar{x}) + \beta_2(x_{2\ell} - \bar{x}) + \beta_3(x_{3\ell} - \bar{x}) + \varepsilon_\ell,$$

where $x_{i\ell} - \bar{x}$ is the centered covariate if observation ℓ is in treatment i and 0 otherwise:

$$\mathbf{X} = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & x_{1j} - \bar{x} & 0 & 0 \\ \hline 1 & 0 & 1 & 0 & x_{2j} - \bar{x} & 0 \\ \hline 1 & -1 & -1 & 0 & 0 & x_{3j} - \bar{x} \end{array}\right],$$

each row block having n rows, and

$$\boldsymbol{\beta} = (\mu, \tau_1, \tau_2, \beta_1, \beta_2, \beta_3)^T.$$

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_a$ (parallel lines). Full model (separate slopes): each treatment is an ordinary simple regression, so

$$SS_E(\mathrm{F}) = \sum_{i=1}^{a} \Big[E_{yy}^{(i)} - \frac{(E_{xy}^{(i)})^2}{E_{xx}^{(i)}}\Big], \quad df_F = a(n-2),$$

with $E^{(i)}$ the within-treatment-i sums. Reduced model = common slope ANACOVA: $SS_E(\mathrm{R}) =$

$E_{yy} - E_{xy}^2/E_{xx}$, $df_R = a(n-1) - 1$. Then

$$F_0 = \frac{[SS_E(\mathrm{R}) - SS_E(\mathrm{F})]/(a-1)}{SS_E(\mathrm{F})/[a(n-2)]} \overset{H_0}{\sim} F_{a-1,\,a(n-2)}.$$

Example 11.6: Parallelism for the cracker data

Within-treatment sums: trt 1: $E_{xx}^{(1)} = 54.8$, $E_{yy}^{(1)} = 78.8$, $E_{xy}^{(1)} = 59.8$; trt 2: 137.2, 142.0, 136.0; trt 3: 141.2, 86.8, 103.6. Separate slopes 1.091, 0.991, 0.734.

$$\begin{aligned} SS_E(\mathrm{F}) &= \left(78.8 - \tfrac{59.8^2}{54.8}\right) + \left(142.0 - \tfrac{136.0^2}{137.2}\right) + \left(86.8 - \tfrac{103.6^2}{141.2}\right) \\ &= 13.54 + 7.19 + 10.79 = 31.52 \ (9 \text{ df}), \\ F_0 &= \frac{(38.57 - 31.52)/2}{31.52/9} = \frac{3.53}{3.50} = 1.01 < F_{0.05,2,9} = 4.26. \end{aligned}$$

No evidence against parallel slopes: the common-slope ANACOVA is appropriate.

Example 11.7: Fiber strength by machine

Three machines, $n = 5$, y = strength, x = thickness:

	Machine 1		Machine 2		Machine 3	
	y	x	y	x	y	x
	36	20	40	22	35	21
	41	25	48	28	37	23
	39	24	39	22	42	26
	42	25	45	30	34	21
	49	32	44	28	32	15
Total	207	126	216	130	180	106

$S_{xx} = 261.73$, $S_{xy} = 282.60$, $S_{yy} = 346.40$; $T_{xx} = 66.13$, $T_{xy} = 96.00$, $T_{yy} = 140.40$; $E_{xx} = 195.60$, $E_{xy} = 186.60$, $E_{yy} = 206.00$.

$$\hat\beta = 186.6/195.6 = 0.954, \quad SS_E(\mathrm{F}) = 206.0 - \tfrac{186.6^2}{195.6} = 27.99 \ (11 \text{ df}), \quad MS_E = 2.544,$$

$$SS_E(\mathrm{R}_\tau) = 346.4 - \tfrac{282.6^2}{261.73} = 41.27, \quad F_0 = \frac{(41.27 - 27.99)/2}{2.544} = 2.61 < 3.98:$$

no machine difference after adjusting for thickness. Slope: $F_0 = \frac{186.6^2/195.6}{2.544} = 69.97 > 4.84$: thickness is significant. $\hat\mu = 40.2$, $\hat\tau = (0.182,\ 1.219,\ -1.402)$. Parallelism: $SS_E(\mathrm{F}) = 25.25$ (9 df), $F_0 = \frac{(27.99-25.25)/2}{25.25/9} = 0.49$: parallel slopes accepted.

7 Assumption Checking

Residuals $e_{ij} = y_{ij} - \hat y_{ij}$: normal Q–Q plot; residuals vs fitted values; residuals vs treatment; residuals vs x (checks linearity in the covariate). For the cracker data all plots are satisfactory.

8 Selected exercises

Motivation. The adjusted treatment means are the quantities reported from an analysis of covariance; the first exercise gives their precision, the second measures exactly how much the covariate

gains over the one-way ANOVA.

Exercise 11.1 (Adjusted treatment means; Montgomery 15.18)

In the model of this chapter, let $\hat{\mu}_i^{\text{adj}} = \bar{y}_{i\cdot} - \hat{\beta}(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})$.
(a) Show $\mathrm{E}\hat{\beta} = \beta$ and $\mathrm{E}\hat{\mu}_i^{\text{adj}} = \mu + \tau_i$.
(b) Show $\mathrm{Cov}(\bar{y}_{i\cdot}, \hat{\beta}) = 0$ and $\mathrm{Var}(\hat{\mu}_i^{\text{adj}}) = \sigma^2\Big[\frac{1}{n} + \frac{(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2}{E_{xx}}\Big]$.
(c) Deduce the $100(1-\alpha)\%$ interval

$$\hat{\mu}_i^{\text{adj}} \pm t_{\alpha/2,\,a(n-1)-1}\sqrt{MS_E\Big[\frac{1}{n} + \frac{(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2}{E_{xx}}\Big]}$$

and evaluate it for promotion 1 of the cracker data ($t_{0.025,11} = 2.201$).

Solution.

(a)

Step 1. $E_{xy} = \sum_i \sum_j (x_{ij} - \bar{x}_{i\cdot})y_{ij}$ because $\sum_j (x_{ij} - \bar{x}_{i\cdot})\bar{y}_{i\cdot} = 0$. Taking expectations, the terms $\mu + \tau_i$ and $-\beta\bar{x}_{\cdot\cdot}$ are constant within treatment i and vanish against $\sum_j (x_{ij} - \bar{x}_{i\cdot}) = 0$, leaving

$$\mathrm{E}E_{xy} = \beta \sum_i \sum_j (x_{ij} - \bar{x}_{i\cdot})x_{ij} = \beta E_{xx},$$

so $\mathrm{E}\hat{\beta} = \beta$.

Step 2. $\mathrm{E}\bar{y}_{i\cdot} = \mu + \tau_i + \beta(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})$, hence $\mathrm{E}\hat{\mu}_i^{\text{adj}} = \mu + \tau_i$.

(b) By independence, $\mathrm{Cov}(\bar{y}_{i\cdot}, \hat{\beta}) = \frac{\sigma^2}{nE_{xx}} \sum_j (x_{ij} - \bar{x}_{i\cdot}) = 0$. Therefore

$$\mathrm{Var}(\hat{\mu}_i^{\text{adj}}) = \mathrm{Var}(\bar{y}_{i\cdot}) + (\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2 \mathrm{Var}(\hat{\beta}) = \frac{\sigma^2}{n} + \frac{\sigma^2(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2}{E_{xx}}.$$

(c)

Step 1. $\hat{\mu}_i^{\text{adj}}$ is a linear function of the least squares estimate, so it is normal with the mean and variance above, and it is independent of SS_E (as in Ch. 8, $\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = \mathbf{0}$), where $SS_E/\sigma^2 \sim \chi^2_{a(n-1)-1}$.

Step 2. Hence $\{\hat{\mu}_i^{\text{adj}} - (\mu + \tau_i)\}/\sqrt{MS_E[\cdots]} \sim t_{a(n-1)-1}$; inverting gives the interval.

Step 3 (cracker, promotion 1). $\hat{\mu}_1^{\text{adj}} = 39.82$, $\bar{x}_{1\cdot} - \bar{x}_{\cdot\cdot} = -1.8$, $MS_E = 3.506$, $E_{xx} = 333.2$:

$$\widehat{\mathrm{Var}} = 3.506\Big(\frac{1}{5} + \frac{3.24}{333.2}\Big) = 0.735, \qquad 39.82 \pm 2.201\sqrt{0.735} = 39.82 \pm 1.89 = [37.93,\ 41.71].$$

The farther a treatment's covariate mean lies from $\bar{x}_{\cdot\cdot}$, the less precise its adjusted mean. ■

Exercise 11.2 (What the covariate buys)

Let $r_w = E_{xy}/\sqrt{E_{xx}E_{yy}}$ be the pooled within-treatment correlation of x and y.
(a) Show $SS_E = (1 - r_w^2)E_{yy}$, so the ratio of the ANCOVA error mean square to the one-way ANOVA error mean square is $(1 - r_w^2)\,a(n-1)/\{a(n-1)-1\}$. Check this against the cracker data.
(b) If the covariate means are equal in all treatments ($T_{xx} = 0$), show that $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ and that the adjusted treatment sum of squares equals T_{yy}.

Solution [Supplement].

(a) $SS_E = E_{yy} - E_{xy}^2/E_{xx} = E_{yy}\{1 - E_{xy}^2/(E_{xx}E_{yy})\} = (1 - r_w^2)E_{yy}$. The one-way ANOVA has error SS E_{yy} on $a(n-1)$ df and ANCOVA has SS_E on $a(n-1)-1$ df, which gives the ratio. Cracker data: $r_w^2 = 299.4^2/(333.2 \times 307.6) = 0.875$, and $(1 - 0.875) \times 12/11 = 0.137 = 3.506/25.63$: one degree of freedom spent on β removes about 87% of the error.

(b)

Step 1. $T_{xx} = n\sum_i(\bar{x}_{i\cdot} - \bar{x}_{\cdot\cdot})^2 = 0$ forces $\bar{x}_{i\cdot} = \bar{x}_{\cdot\cdot}$ for all i, so $\hat{\tau}_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ and also $T_{xy} = 0$.

Step 2. Then $S_{xx} = E_{xx}$ and $S_{xy} = E_{xy}$, so the adjusted treatment SS is

$$\left(S_{yy} - \frac{S_{xy}^2}{S_{xx}}\right) - \left(E_{yy} - \frac{E_{xy}^2}{E_{xx}}\right) = S_{yy} - E_{yy} = T_{yy}.$$

With balanced covariate means the treatment comparison is unchanged and the covariate acts only on the error, exactly like blocking; when the $\bar{x}_{i\cdot}$ differ, the adjustment also corrects the treatment means, at the cost of the extra variance in Exercise 11.1(b). ■